

Amar Dhillon

NYC-Metropolitan Area | amardhillon844@gmail.com | (XXX) XXX-XXXX | linkedin.com/in/amar-dhillon-917537261 | github.com/AmarDhillon05

EDUCATION

University of Maryland, College Park
B.S. in Computer Science, Minor in Statistics; GPA: 3.5

College Park, MD
May 2028

CERTIFICATIONS

Amazon Web Services (AWS) Certified Developer - Associate
Amazon Web Services

Nov 2024 – Nov 2027
Digital Badge

EXPERIENCE

Software Development Engineer Intern
Amazon Web Services

Sep 2026 – Dec 2026
Redmond, WA

- Researching and implementing Java-based capacity optimizations for Amazon Redshift, backtesting upper and lower bounds on the EC2 capacity reserved for patching to balance availability against cost

Software Engineering Intern
Capital One

Jun 2026 – Aug 2026
Richmond, VA

- Overhauled the legacy UI of a test data workflow creator using React + Electron with TypeScript workflow validation backed by Jest testing and a React-Flow editor, preferred over the legacy tool by **90%** of 30+ surveyed developers
- Engineered a precomputed tree-style workflow search index as the Claude skill's main search tool, distilling popular workflows from a **270K+** row PostgreSQL table with Python + SQLAlchemy and Redis caching to stay within database permission and rate limits
- Developed a Claude skill for workflow generation using sub-skill decomposition and Python functions for workflow search and validation, guided by CloudWatch + OpenSearch latency logs, cutting generation time **65%** to under 15 seconds

Software Engineer (Contract)
Amazon (via UMD App Development Contracting)

Jan 2026 – Present
College Park, MD

- Deployed an Amazon Leo topic-scoring research utility that shows public-policy staff which space topics are popular, through Docker + AWS Fargate and GitHub Actions CI/CD with AWS CDK (TypeScript) production infrastructure
- Achieved a **34%** end-to-end latency reduction on a pipeline routing 6,000+ scraped research documents for inference by replacing S3 staging hops with Kinesis streams feeding Lambda processors, with API Gateway fronting for app integration
- Integrated production fine-tuning of an AWS Bedrock document-importance model through EventBridge cron triggers with an LLM-as-judge decision gate to avoid redundant retraining, with X-Ray tracing and SNS failure alerts

Machine Learning Research Intern
UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present
College Park, MD

- Developing VLM-based robot grasp selection for PhysTwin, a physics-based digital twin, with a PhD research team targeting ICLR
- Built a Python pipeline linking AnyGrasp multi-camera grasp candidates to Open3D overlay views consumed by a Qwen3-VL-8B VLM, isolating scene components, and tuned its prompts to pick grasps that avoid hitting the table
- Cut research-environment setup time **30%** via uv dependency pinning and Docker, with psutil timing and memory monitoring and Pytest unit testing for identifying bottlenecks

Co-Founder, Software Engineer
TerpLabs

Jun 2025 – Present
College Park, MD

- Co-founded and scaled a **110+ member** campus product studio shipping 5 live products, organizing teams with Jira and GitHub through agile practices
- Sustained **20,000+ visits** without crashing in the first 30 days of Tortuga, a class scheduler, through rewriting Prisma-based SQL queries with Redis query caching to take load off compute
- Building a RAG meal suggestion feature for the official UMD app using GPT-4o mini over ChromaDB + SentenceBERT vector search on scraped recipes, achieving **93%** meal satisfaction on internal test users

PROJECTS

IEX Order Book & Matching Engine | *C++17, STL, POSIX, Linux perf*

Jul 2026 – Aug 2026

- Implemented a C++17 order book and matching engine on historical IEX data, replaying **50.9M** real exchange messages with no resync and matching 99,122 of 99,123 fills and 100% of top-of-book prices
- Designed the order book with std::map price levels of FIFO queues and an order-ID hash map for O(1) cancels, sustaining **1.26M** events/sec end-to-end over the full replay
- Designed and benchmarked LSM-tree and B+ tree storage for simulation logs on identical 2M-record workloads, with the B+ tree giving **12.8x** read throughput and 22x faster range queries for 27% more disk

TECHNICAL SKILLS

Languages: Python, Java, C++, C, TypeScript/JavaScript, SQL

AWS: Lambda, Kinesis, Fargate/ECS, API Gateway, EventBridge, Bedrock, CDK, CloudWatch, X-Ray, SNS, IAM

Backend & Data: Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

AI/ML & Tools: PyTorch, Hugging Face, RAG, Open3D, Docker, GitHub Actions, Linux, Pytest, Jest, uv, perf